


CLOUDFLIGHT
CODING
CONTEST

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# Asteroids might be rotated



- **For each asteroid, dig tunnels that discover every cell**
  - "discover every cell" is explained on the next page
- Either the **width or the height** of the minable rectangle is guaranteed to be 3
- The other dimension is any integer between 3 and 199 (inclusive)
- The outpost is now **anywhere in the first row** of the asteroid
- A dig limit is given
- Your tunnels must not have more dug cells than the dig limit
- The given dig limit requires an **optimal solution**

**There is a visualizer.html in the input folder.**
**You can use it to visualize your solutions.**

# Valid Tunnel Layout Checklist

- **Discover every cell:** Each cell inside the minable rectangle must either be a tunnel, or be adjacent to a tunnel
- Your tunnels may branch at any point, but must be connected to the outpost as well as to each other
- Diagonals do not count as adjacent or connected
- Bedrock cannot be dug
- Your tunnels must not have more cells than the dig limit



# Input

| Name | | Description | Example |
|---|---|---|---|
| N | | Number of asteroids | 2<br>3 5 6<br>#S###<br>#:::#<br>#:::#<br>#:::#<br>#:::#<br>#:::#<br>#####<br><br>5 3 6<br>####S##<br>#:::::#<br>#:::::#<br>#:::::#<br>####### |
| repeated N times | Minable rectangle size and **dig limit** | The width and height of the minable rectangle followed by the dig limit, separated by a spaces | |
| | Character representation | A paragraph of characters<br>‘#’ represents bedrock<br>‘:’ represents a minable cell<br>‘S’ represents the outpost | |
| | Empty line between asteroids | | |

# Output

| Name | Description | Example |
|---|---|---|
| Character representation (repeated N times) | A paragraph of characters<br>‘#’ represents bedrock<br>‘:’ represents a minable cell<br>‘S’ represents the outpost<br>‘X’ represents a tunnel cell | #S###<br>#XX:#<br>#:X:#<br>#:X:#<br>#:X:#<br>#:X:#<br>#####<br><br>####S##<br>#:::X:#<br>#XXXXX#<br>#:::::#<br>####### |
| *Mandatory empty line between asteroids* | | |




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